

Sebastian Laverde Alfonso

[Agentic AI Systems](#) · [LLM Infrastructure](#) · [Document Intelligence](#) · [Advanced RAG](#)

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Machine Learning Engineer with 7+ years of experience across the full ML lifecycle, from research and fine-tuning through production deployment, observability, and scale. Built and shipped deep learning systems for document intelligence, vision-language understanding, NLP, and multimodal AI across 8+ industries and 15+ shipped projects, with deep specialization in agentic AI architectures and LLM infrastructure.

CORE COMPETENCIES

Agentic AI and Orchestration: LangGraph, LangChain, CrewAI, AutoGen, LlamaIndex; ReAct, Plan-and-Solve, A2A protocol design, MCP.

RAG and Information Retrieval: Hybrid search (dense + sparse / SPLADE++ / RRF), Qdrant, FAISS, Pinecone, Weaviate; embedding models, semantic search, cross-encoder reranking, document chunking strategies, caching, multi-tenant retrieval.

LLM Engineering and Fine-tuning: Parameter-efficient fine-tuning (LoRA, QLoRA), RLHF / DPO; prompt engineering, LLM evaluation and benchmarking, multi-provider strategy (Vertex AI / Gemini, Anthropic Claude, OpenAI), model quantisation (ONNX, INT8).

Applied ML and Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost; transformer architectures (BERT, Donut, T5), vision-language models, multimodal learning, zero-shot and few-shot learning, active learning, synthetic data generation, VAE.

Production AI and MLOps: FastAPI, Docker, CI/CD; MLflow, DVC, Weights and Biases; data versioning, ETL automation, model monitoring, A/B testing, LLM evaluation pipelines (LangSmith), structured logging, latency SLOs

Observability: OpenTelemetry, LangSmith, Traceloop, GCP Cloud Trace; distributed tracing, LLM observability, monitoring and alerting.

Cloud | Infrastructure: GCP (Vertex AI, Cloud Run, Firestore), AWS (SageMaker, Lambda, EC2, S3), Azure, Kubernetes.

Programming and Tools: Python (expert, 3.12+), SQL, Bash, Git, uv, Pydantic.

Languages: Spanish (native), English (fluent), German (B1+)

WORK EXPERIENCE

Sapience AI Corporation *Nov 2025 – Apr 2026*

Fullstack AI Engineer - Sole Architect and Technical Owner

Sole architect and technical owner of a production multi-agent AI SaaS platform, designing four autonomous agents (CRM intelligence, enterprise RAG, corporate research, social media), inter-agent orchestration, and cloud infrastructure on GCP Cloud Run.

- **Designed** a custom Agent-to-Agent (A2A) orchestration framework enabling secure, multi-tenant, independently deployable AI services.
- **Built** enterprise RAG infrastructure including multi-format document ingestion (PDF, Office, email, OCR), hybrid retrieval, and optimized vector indexing achieving 72% storage reduction..
- **Established** end-to-end production observability: OpenTelemetry + GCP Cloud Trace + Traceloop SDK for LLM-specific span capture (latency); LangSmith for evaluation and prompt versioning; structured Cloud Logging; latency SLO alerting.
- **Owned** production GCP infrastructure, CI/CD, security, state management, and deployment lifecycle for all AI services.
- **Mentored** the engineering team on LangGraph patterns, production error handling, and LLM observability; established an extensible tool registry pattern adopted across all agents.

Unstructured Technologies (unstructured.io) *Jan 2022 - Dec 2024*

Data Scientist / ML Engineer

- **Fine-tuned** transformer-based vision-language models on proprietary document datasets to parse diverse layouts into structured JSON; achieved 90%+ accuracy on invoices | receipts | forms, improving client satisfaction with the flagship pipeline.
- **Led** the ML effort to train lightweight classifiers (XGBoost + sentence embeddings + NLP features) for key-value extraction from complex form images; increased extraction accuracy from a 40% baseline to 76% on standard layouts through iterative feature engineering and data-driven model selection.

- **Designed** chunking strategies, evaluation frameworks, prompt engineering, and metadata enrichment pipelines for financial RAG systems, improving retrieval and structured extraction quality across diverse documents.
- **Co-authored** peer-reviewed publication: "[Financial Report Chunking for Effective Retrieval Augmented Generation](#)" ([arXiv:2402.05131, 2024](#)) documenting advanced chunking strategies for financial document RAG systems.

German Aerospace Centre (DLR), Security Department *Feb 2021 - Oct 2021*

Research Assistant, Deep Learning

- **Designed** and delivered a Dockerized conditional VAE (with holistic discriminators) for medical record de-identification, synthesizing privacy-compliant records conditioned to patient profiles, enabling downstream clinical NLP research in a high-security environment. Publication: [Clinical NLP with Controlled Synthetic Medical Record Generation \(PDF\)](#).

Webis Group, Bauhaus-Universität Weimar *Oct 2018 - Apr 2022*

Information Retrieval and Deep Learning Research Assistant

- **Developed** transformer-based NLP systems for multilingual zero-shot text classification, code generation, and argument search, achieving **90%+ accuracy across 100 categories** and **70% accuracy** in political inclination prediction.

Omdena, Collaborative AI Platform *Oct 2018 - 2020*

Machine Learning Engineer and Community Builder

- **Built** ML solutions for social good, including NLP classifiers for sexual harassment report categorization, renewable energy demand prediction, and Mars anomaly detection, while mentoring AI education initiatives and publishing technical research and open-source contributions.

Data Analysis Consultant / Freelancer *Nov 2012 - Oct 2017*

Data Analyst and BI Automation Specialist

- **Automated** BI reporting and designed real-time KPI dashboards across multiple client verticals (2012–2017), providing the data engineering foundation for later ML work.

P U B L I C A T I O N S

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- **Jimeno Yepes A., You Y., Milczek J., Laverde S., Li L.** [Financial Report Chunking for Effective Retrieval Augmented Generation](#). [arXiv:2402.05131 \(2024\)](#). [arxiv.org/abs/2402.05131](#)
 - **Laverde Alfonso, S.** [Clinical NLP with Controlled Synthetic Medical Record Generation \(PDF\)](#). Master Thesis, Bauhaus-Universität Weimar / Webis Group (2022). PDF available via [downloads-cf.webis.de](#).
 - **Portfolio:** [sebastian-laverde-ml-portfolio.vercel.app](#).

E D U C A T I O N

MSc, Computer Science for Digital Media - Bauhaus-Universität Weimar, Germany *2018 - 2022*

BSc, Electronic Engineering - Pontifical Xavierian University, Bogota, Colombia *2009 - 2017*

Courses and Certifications at [linkedin.com/in/sebastian-laverde-alfonso](#)